

Turning data into innovative and impactful solutions

DATA SCIENCE PORTFOLIO

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diBimbing.id



Halo!

I'm an English Literature and Data Analytics Graduate passionate about leveraging data to solve complex and impactful problems.

Focus Areas: Data Analytics | Data Science | Machine Learning

- 🛠 Seeking opportunities to apply data skills in real-world business and technical challenges
- ⚡ Consistently sharpening problem-solving skills through coding platforms and self-learning
- 🌱 Highly motivated by challenging problems that require structured thinking and data-driven insights





Education

Data Science & Data Analyst Bootcamp — Dibimbing.id (2026)

- Hands-on experience in data cleaning, EDA, and data visualization
- Developed business dashboards using Power BI
- Applied SQL and machine learning for real-world data problems

Bachelor of Arts — Gunadarma University (2013 – 2017)

- GPA: 3.32

Skills

Tools & Technologies



Analytical Thinking

Ability to analyze and interpret data to generate meaningful and actionable insights



Problem Solving

Capable of solving complex problems using a structured and data-driven approach.



Programming

Python (Pandas, NumPy, Scikit-learn)



Data Analysis

SQL, Microsoft Excel



Attention to Detail

Ensuring data accuracy and consistency to produce reliable analysis



Data Communication

Able to present data insights in a clear and understandable way for decision-making



Data Visualization

Power BI



Machine Learning

Classification Models, Model Evaluation

Transforming data into insights that drive real-world impact.

Work Experience



diBimbing.id



Credit Analyst — Akulaku

Improved loan approval accuracy (+15%) & reduced delinquency (-10%)



Consultant — Solid Gold Berjangka

Delivered data-driven insights to improve client retention (+15%)



Archive Staff — ANRI

Managed 10,000+ records ensuring data accuracy & compliance

Certification



RevoU Mini Course Intro to Data Analytics



diBimbing.id Data Science Bootcamp

Portofolio Projects



Projek 1

Restaurant Tip Analysis

Projek 2

**Customer Behavior & Churn
Prediction**

Projek 3

**Optimasi Funnel Konversi &
Strategi Retensi Pelanggan
E-Commerce.**

Project 1

Restaurant Tip Analysis

Unlocking the Drivers Behind Customer Tips



Restaurant Tip Analysis: Data-Driven Service Optimization

- **The Challenge:** Inconsistent tipping patterns make it difficult to measure service quality and predict staff financial well-being.
- **The Solution:** Built a predictive model to identify key variables that drive tip amounts and optimize restaurant operations.
- **Key Results:** Achieved an R-Squared of 0.52, successfully proving that bill size and timing are the primary predictors of tips.
- **Critical Insight:** While large groups have higher bills, groups of 2 provide the most stable tip percentages; Dinner and Weekends are the peak periods for high-value tips.
- **Business Impact:** Increased Revenue per Table through targeted upselling and improved staff retention by optimizing high-tip shift scheduling.

Project 2

Customer Behavior & Churn Prediction

Predicting Customer Churn Before It Happens



Telco Churn Prediction: Proactive Revenue Recovery

- **The Challenge:** High-value customers (\$74/mo) churning 21% more than loyal users, causing significant revenue leakage.
- **The Solution:** Developed an Early Warning System using a predictive model to identify churn risk before it happens.
- **Key Results:** Achieved 90% Recall and 0.95 AUC, successfully detecting 9 out of 10 potential churners.
- **Critical Insight:** Customer dissatisfaction peaks in the first 6 months; adding a 3rd service reduces churn risk by 50%.
- **Business Impact:** Potential recovery of \$241,200 in annual revenue through targeted retention strategies.

Project 3

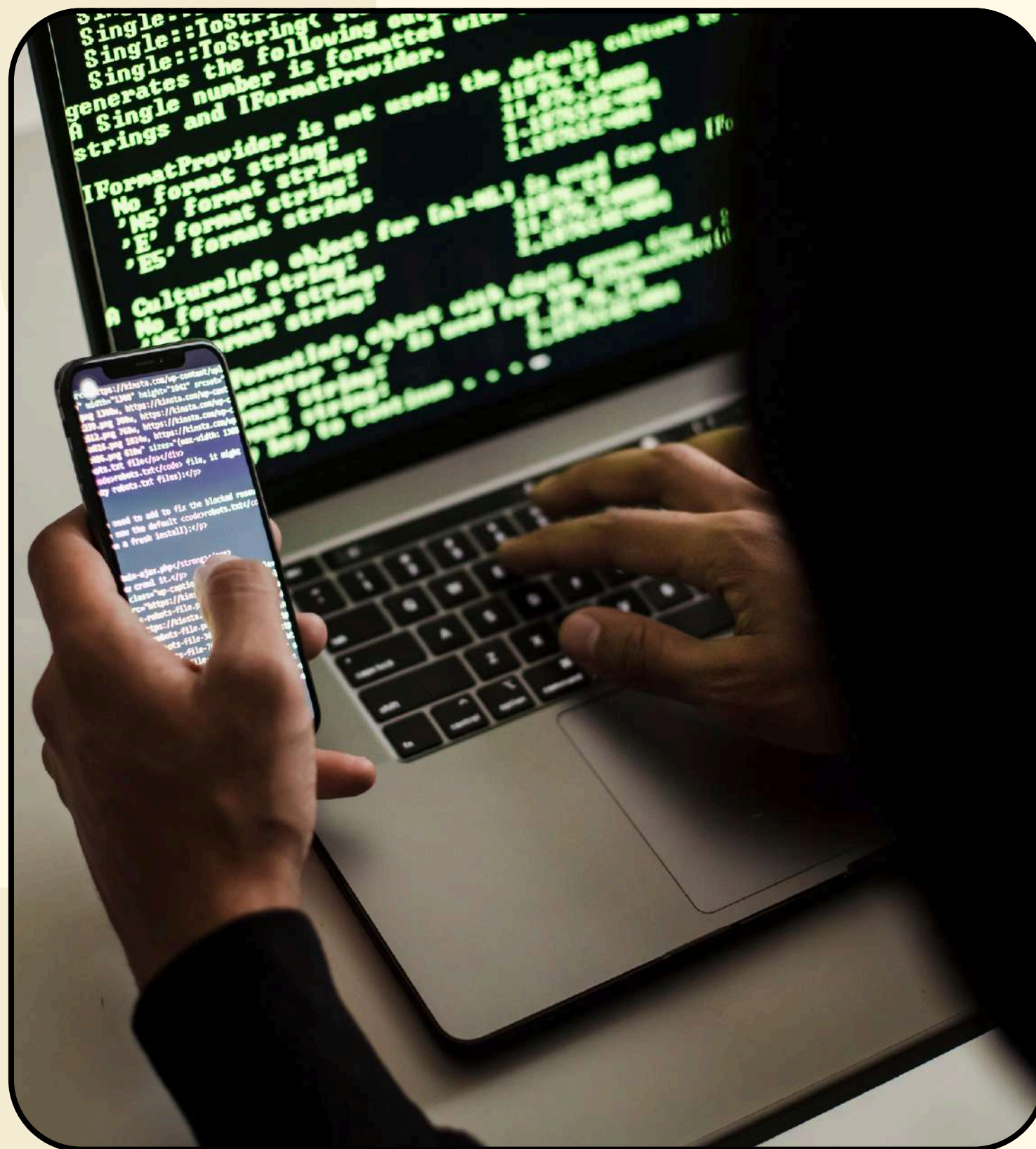
Optimasi Funnel Konversi & Strategi Retensi Pelanggan E-Commerce.

Turning Funnel Data into Conversion Growth



E-Commerce Conversion Optimization & Retention Strategy

- **The Challenge:** High traffic volume but low conversion rates due to significant customer drop-offs at the Product Detail Page (PDP) stage.
- **The Solution:** Conducted a comprehensive Funnel Analysis and RFM (Recency, Frequency, Monetary) segmentation to identify friction points and high-value user behavior.
- **Key Results:** Built an Integrated Performance Dashboard that provides real-time visibility into conversion KPIs and identifies "Power Users" for retention.
- **Critical Insight:** The Product Detail Page is the primary bottleneck for revenue; customer activity peaks at specific hours, offering a window for high-ROAS marketing.
- **Business Impact:** Increased conversion efficiency by fixing funnel leaks and boosted customer lifetime value (LTV) through loyalty programs targeting high-frequency segments.



Fraud Detection System using Machine Learning Real-Time Risk Detection & Cost Optimization

Stopping Fraud Before It Happens

A Data-Driven Approach to Real-Time Risk Detection and Cost Optimization

Executive Summary & Business Problem



Real-Time Fraud Detection with Measurable Cost Impact

Fraud in digital transactions is increasingly sophisticated and difficult to detect due to its extreme rarity compared to legitimate activity. This project develops a real-time fraud detection system using machine learning, designed to minimize false negatives (undetected fraud) while preserving a seamless experience for legitimate users.

The model successfully detected **89% of fraudulent transactions** and improved operational cost efficiency by approximately **24%**, demonstrating its effectiveness in balancing risk mitigation and business performance.

Problem Statement



Fraud Monitoring Project



Imbalanced Data

Fraud transactions are extremely rare (~0.17%) but have massive financial impact on institutions and customers alike.



System Limitations

Traditional rule-based systems fail to detect complex, evolving fraud patterns that adapt over time.



False Negative Risk

Missing a fraud case (false negative) results in direct financial loss — far more costly than a false alarm.

Objective: Build a real-time fraud detection model that maximizes Recall without significantly sacrificing Precision.

Dataset Overview

284,807

Total Transactions

492

Fraud Cases

0.17%

Fraud Rate

30

Features (V1–V28 + Time, Amount, Class)

Features

- Time — seconds from first transaction
- Amount — transaction amount (USD)
- V1–V28 — PCA-transformed confidential features
- Class — target: 0 = Normal, 1 = Fraud

Data Source

European cardholders (September 2013)

Source: Kaggle — Credit Card Fraud Detection Dataset

Highly imbalanced — requires special handling strategy to prevent model bias toward majority class.

Data Preprocessing

01

Standardization

Scaled Time & Amount features using StandardScaler for uniform distribution and improved model convergence.

02

No Undersampling

Preserved original data distribution to avoid information loss. Imbalance handled through model-level strategies.

03

Imbalance Handling

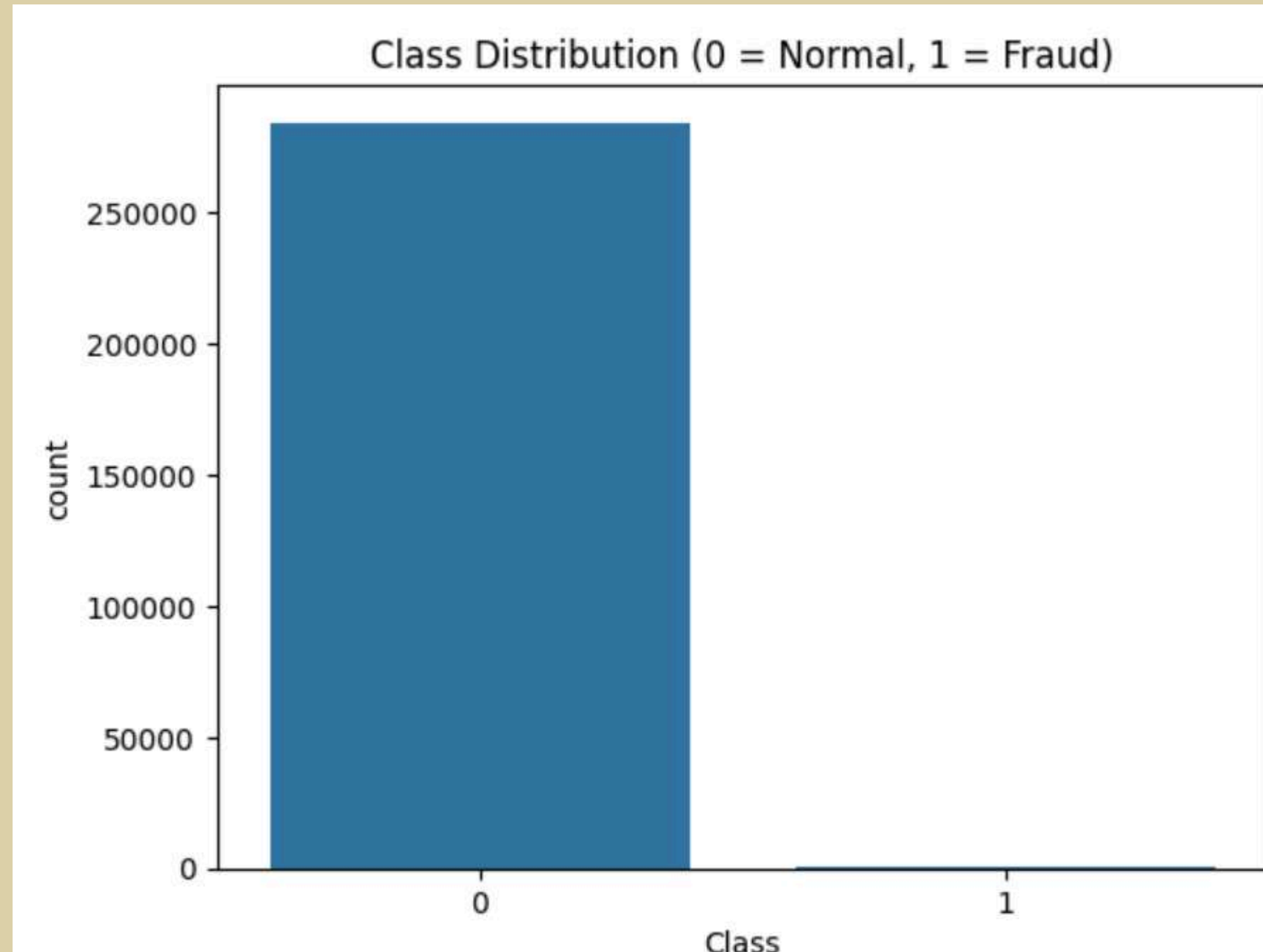
Addressed class imbalance via class weighting and threshold tuning rather than resampling techniques.

04

Train-Test Split

Stratified split to maintain fraud ratio in both training and test sets, ensuring representative evaluation.

Class Distribution



The Insight:

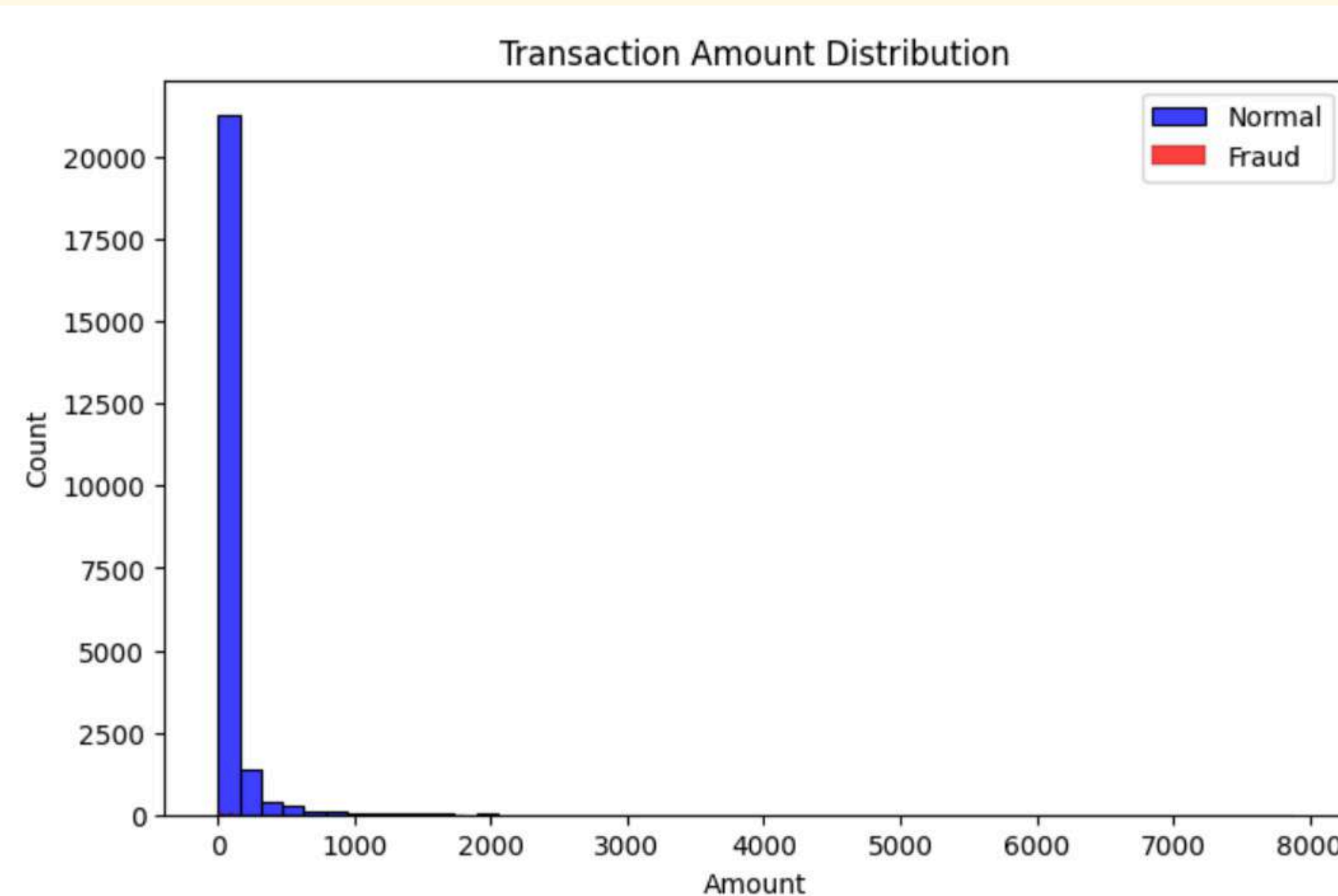
Extreme class imbalance (>99% normal vs ~0.17% fraud) makes accuracy a misleading metric, as models can appear highly accurate while failing to detect actual fraud. To address this, a non-conventional approach is applied by preserving the original data distribution and leveraging robust algorithms (XGBoost) alongside precision-recall focused evaluation metrics, ensuring effective fraud detection without compromising customer experience.

The "Hidden Signals"

Correlation Between Normal vs Fraud Transaction

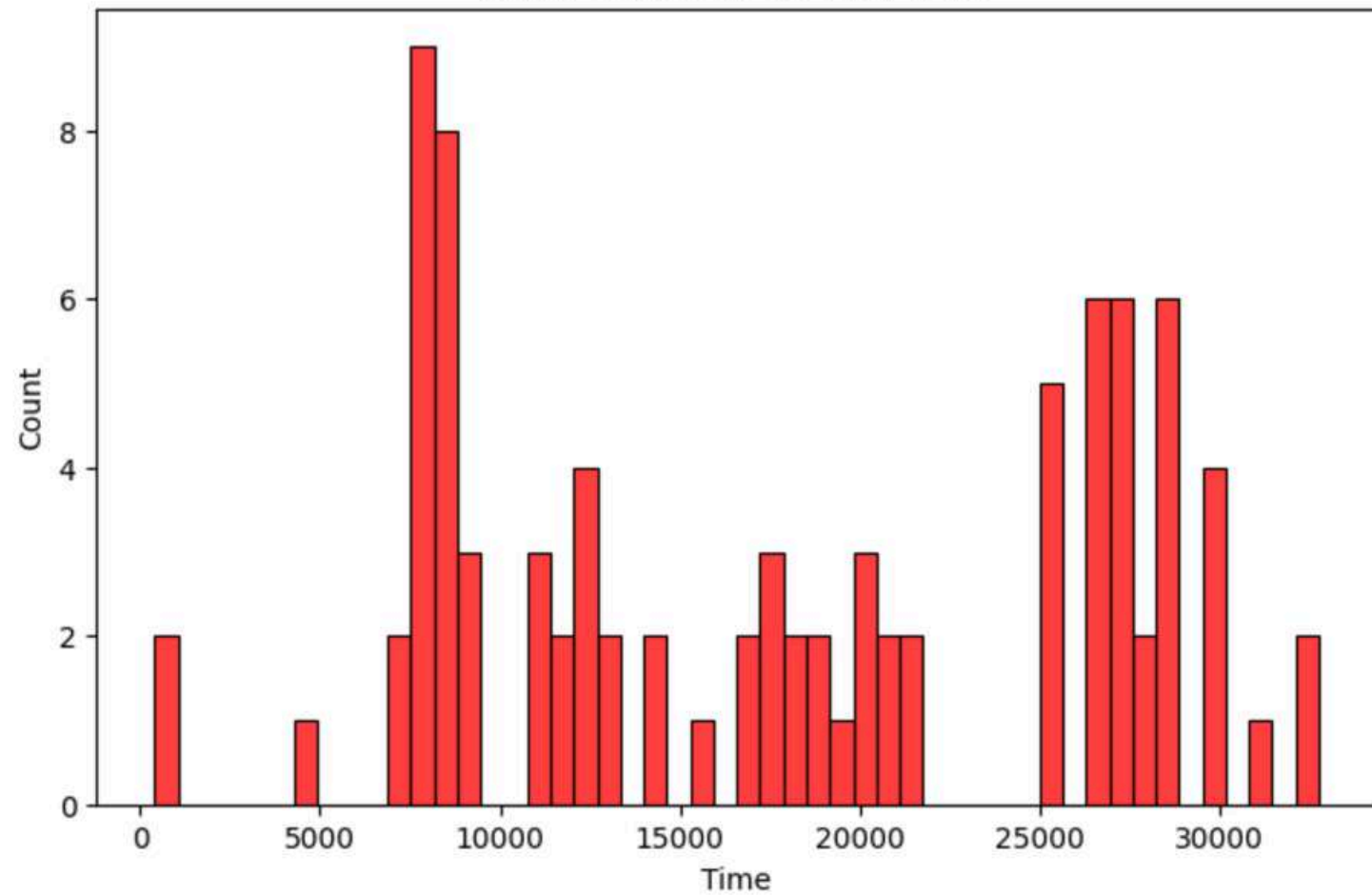
Insight

Legitimate transactions are heavily skewed toward lower values (median ~\$20–\$30) with a long-tail distribution, while fraudulent transactions cluster within a narrower, mid-range band (~\$100–\$200). This pattern suggests that fraudsters deliberately operate within less suspicious ranges to evade detection, often using smaller transactions to validate card activity before executing larger attacks.



Time Pattern Analysis

Fraud Transactions Over Time



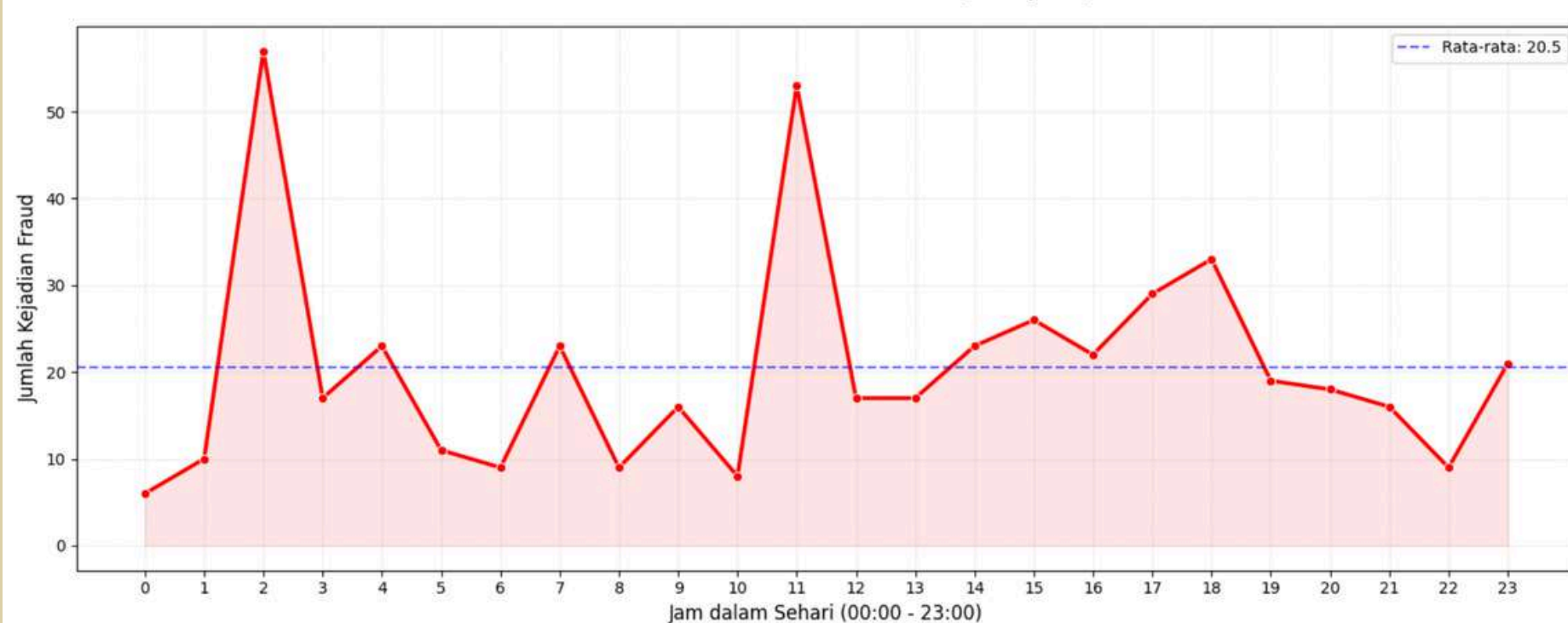
The Insight:

Fraudulent transactions occur consistently across all hours with no significant drop during off-peak periods, unlike normal activity. This indicates that fraud is opportunistic and continuous, making manual monitoring ineffective and reinforcing the need for a real-time, always-on detection system.

Short supporting line:

Fraud activity shows irregular spikes, indicating opportunistic and unpredictable attack patterns.

Tren Waktu Transaksi Fraud (Per Jam)



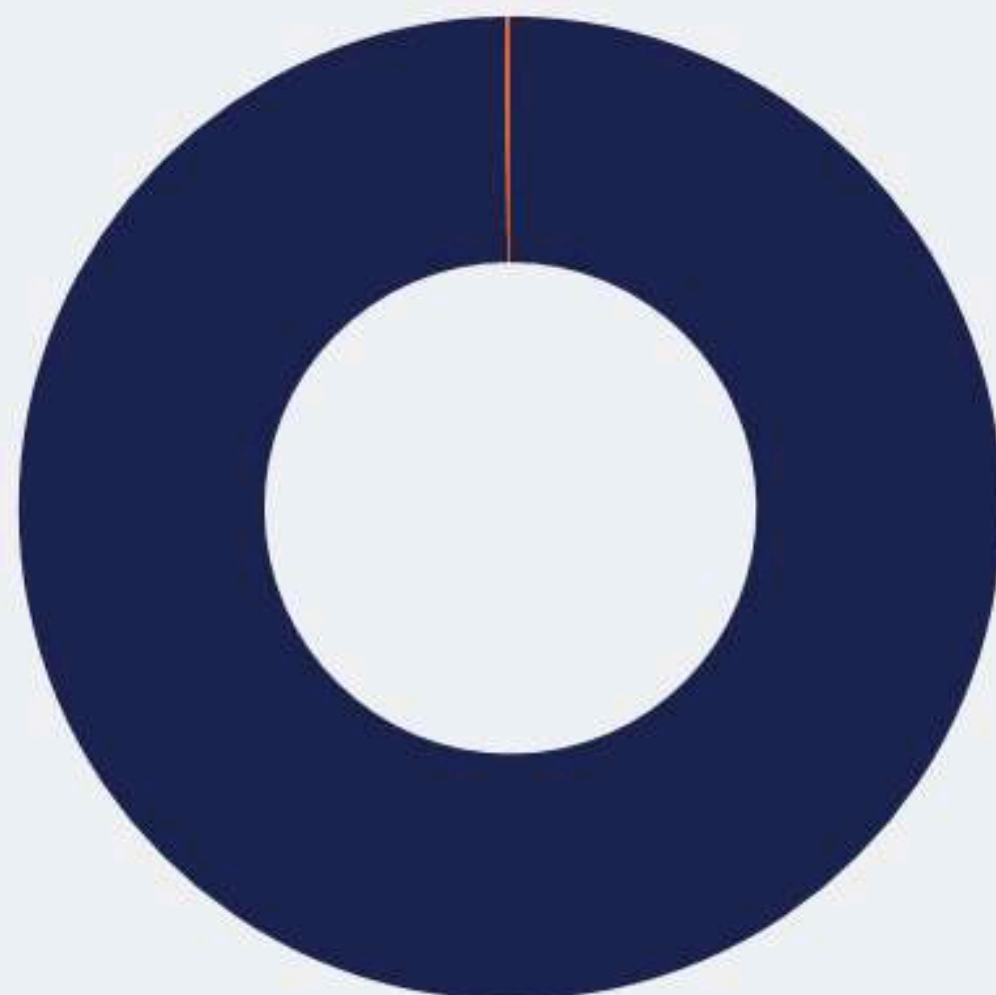
Implication:

Manual monitoring creates blind spots during low-activity periods.

EDA Insights

Uncovering Hidden Patterns Behind Fraudulent Transactions

Class Distribution



■ Normal (99.83%) ■ Fraud (0.17%)



Small Transactions

Fraud predominantly occurs in low-value transactions, suggesting card testing behavior before larger attacks.



24/7 Risk Pattern

Fraud patterns show no time dependency, indicating automated bots operating around the clock.



Hidden in Normal

Fraud distribution overlaps significantly with normal transactions, requiring non-linear detection models.

Modeling Approach

01

Logistic Regression

Baseline Model

- ✓ Fast & interpretable
- ✓ Establishes performance baseline
- ✓ Linear decision boundary

02

Random Forest

Ensemble Model

- ✓ Handles non-linearity
- ✓ Feature importance insights
- ✓ Robust to outliers

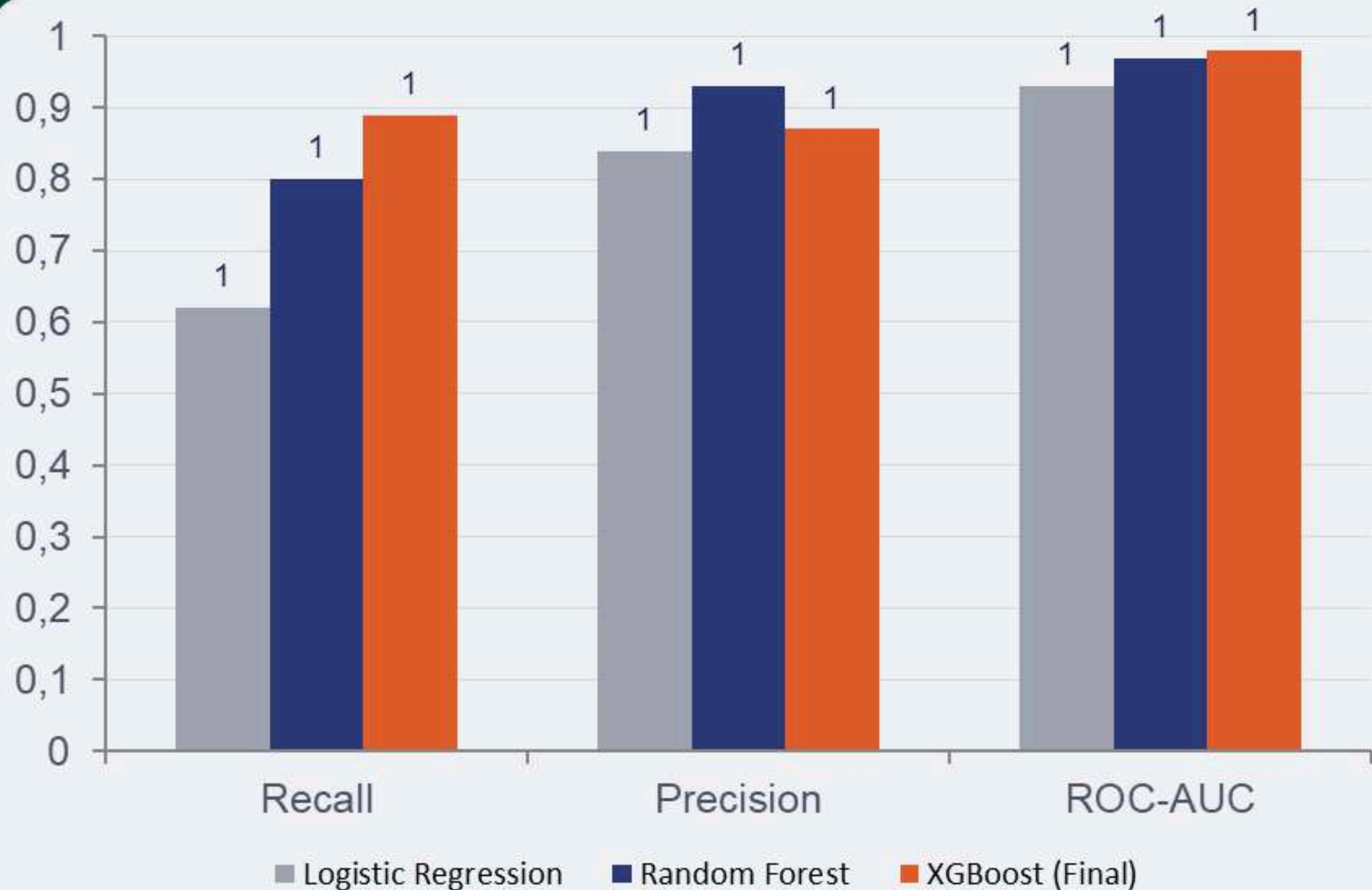
03

XGBoost

Final Model ★

- ✓ Best overall performance
- ✓ Gradient boosting framework
- ✓ Optimized for imbalance

Model Performance



Recall

0.89

Catches 89% of fraud

Precision

0.87

Low false alarms

ROC-AUC

0.98

Near-perfect separation

Key Insights



Behavior Over Amount

Fraud is not determined by transaction size — it's about behavioral patterns. Small amounts can be just as suspicious as large ones.



Critical Features: V14 & V4

V14 and V4 (PCA-transformed) emerged as the most predictive features in distinguishing fraudulent from legitimate transactions.



Non-Linear & Hidden

Fraud patterns are non-linear and deeply embedded in transaction sequences, making linear models insufficient.

Business Impact



89%

Fraud Detected

Minimizes financial losses by catching nearly
9 in 10 fraud cases



~24%

Operational Efficiency

Reduction in manual review workload
through automated detection



Low FP

User Experience

Minimal false positives ensure legitimate
customers are not disrupted

Recommendation & Next Steps

01

Real-Time Detection System

Deploy XGBoost model as a real-time API integrated into the payment gateway pipeline for instant fraud scoring.

02

Early Warning System

Leverage V14 & V4 feature importance to build lightweight early-warning triggers before full model inference.

03

Periodic Retraining

Schedule model retraining quarterly — fraud patterns evolve dynamically, and static models degrade over time.

04

Explainability Layer

Add SHAP values visualization to improve trust, provide audit trails, and enable business interpretation of predictions.

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